

LiDAR vs Radar comparison



i want reason to prove lidar is better than radar

Reasons to prove LiDAR is better than RADAR (for perception and 3D understanding):

1. Much higher spatial resolution

- LiDAR emits millions of narrow laser beams and measures precise time-of-flight.
- This produces **dense 3D point clouds** with centimeter-level accuracy.
- Radar has wide beams → sparse points → poor shape detail.

2. Accurate 3D geometry and object shape

- LiDAR directly gives (x, y, z) coordinates of surfaces.
- Enables accurate **object boundaries, size, orientation, and contours**.
- Radar mostly gives range + velocity; shape reconstruction is weak.

3. Superior localization and mapping (SLAM)

- LiDAR-based SLAM works reliably because of rich geometric features.
- Radar maps are noisy and ambiguous due to multipath reflections.

4. Better object classification

- Fine-grained point clouds allow distinguishing **pedestrian vs cyclist vs car**.
- Radar struggles to separate objects with similar radar cross-sections.

5. Lower angular uncertainty

- LiDAR angular resolution is very high (small beam divergence).
- Radar has high angular uncertainty → object positions blur sideways.

6. Direct height information

- LiDAR measures elevation explicitly (bridges, curbs, road slope).
- Radar height estimation is unreliable without complex antenna arrays.

7. Cleaner data for deep learning

- LiDAR point clouds are structured and consistent.
- Radar data is noisy, speckled, and harder to learn from.

8. Better short-mid range perception (0–100 m)

- LiDAR excels at near and mid-range obstacle detection.
- Radar is mainly strong only at long range and velocity estimation.

Where RADAR is better (important to mention for fairness):

- Works in fog, rain, snow
- Measures velocity directly (Doppler)
- Longer range and cheaper

Conclusion:

LiDAR is **better than radar for accurate 3D perception, geometry, mapping, and object understanding**, while radar is complementary for robustness and velocity. That is why modern autonomous systems treat **LiDAR as the primary geometry sensor and radar as a supporting sensor**.

You said:

i want reason to prove lidar is better than radar

Correct version:

I want reasons to prove that LiDAR is better than radar.



does radar finds the object even if the car is hidden by another car?

Yes, radar can sometimes detect an object even if it is partially hidden by another car — but with important limitations.

Why radar *can* see hidden objects

1. Longer wavelength (cm-mm range)

- Radar waves are much longer than LiDAR's laser wavelength.
- Because of this, radar waves can **diffract (bend)** and **reflect** around objects.

2. Multipath reflections

- Radar signals can bounce off the road, buildings, or other vehicles.
- These indirect paths can return a signal from a **partially occluded vehicle**.

3. Strong reflection from metal

- Cars are metallic → high radar cross section (RCS).
- Even a small exposed portion can produce a detectable return.

4. Doppler effect (motion detection)

- Radar detects **velocity directly**.
 - If a hidden car is moving, radar may detect motion even when geometry is unclear.
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Why this detection is unreliable

1. No clear shape or outline

- Radar may detect *something*, but not its exact size or position.

2. Ghost objects

- Multipath causes false positives (phantom cars).

3. Poor angular resolution

- Radar often cannot tell *where exactly* the hidden object is laterally.

4. Complete occlusion = no detection

- If the front car fully blocks reflections and no multipath exists, radar fails.

Comparison summary

Sensor	Hidden object detection
Radar	Sometimes (via diffraction & multipath)
LiDAR	No (line-of-sight only)
Camera	No (line-of-sight only)

Key takeaway

Radar **does not truly “see through” cars**, but it can **sometimes infer the presence of a hidden moving object** due to physics of radio waves. This is exactly why radar is valuable as a **safety backup sensor**, while LiDAR remains superior for **precise geometry**.

You said:

does radar finds the object even if the car is hidden by another car?

Correct version:

Does radar detect an object even if a car is hidden behind another car?



if i ask you lidar or radar for adas which one will you choose and why

I would choose LiDAR for ADAS — with radar as a supporting sensor.

Why LiDAR is the primary choice

1. Precise 3D perception

- ADAS needs to know *where exactly* an object is, not just that it exists.
- LiDAR gives accurate (x, y, z) geometry, enabling reliable obstacle distance, width, height, and shape estimation.

2. Reliable object detection and classification

- Pedestrians, cyclists, vehicles, curbs, road edges — all are clearly separable in LiDAR point clouds.
- Radar often merges multiple objects into one or produces ambiguous detections.

3. Safer decision-making

- Braking, lane changes, and collision avoidance require **precise spatial reasoning**.
- LiDAR reduces false positives and false negatives compared to radar-only systems.

4. Better short-mid range performance

- Most ADAS actions happen within **0–80 m**.
- LiDAR excels exactly in this range, where accuracy matters most.

5. Cleaner input for learning-based systems

- Deep learning models trained on LiDAR converge faster and behave more predictably than radar-based models.

Why radar alone is not enough

- Poor angular resolution
- No reliable object shape
- Ghost targets due to multipath
- Hard to localize obstacles precisely

Radar answers “**something is moving**”, not “**what and exactly where**.”

Balanced engineering truth

If the question were “**one sensor only**”, I choose **LiDAR**.

If the question were “**best real-world ADAS**”, I choose **LiDAR + radar + camera**.

- **LiDAR** → geometry & safety
- **Radar** → velocity & robustness
- **Camera** → semantics & traffic rules

One-line conclusion

LiDAR is chosen because ADAS is a geometry-and-safety problem, and LiDAR is the best sensor for geometry.

You said:

if i ask you lidar or radar for adas which one will you choose and why

Correct version:

If I ask you to choose between LiDAR and radar for ADAS, which one would you choose and why?



* Poor angular resolution * No reliable object shape * Ghost targets due to multipath * Hard to localize obstacles precisely
explain these

Explanation of each point (step by step, beginner-friendly):

1. Poor angular resolution

What “angular resolution” means

- It is the ability of a sensor to distinguish **two objects that are close side-by-side** (left–right direction).

Radar case

- Radar sends **wide radio beams**.
- If two cars are close in angle, radar receives their echoes **as one blob**.

Result

- Radar knows *something is ahead*,
- but cannot tell **exactly where sideways**.

Example

- Two bikes riding side by side → radar may detect **one object**, not two.
-

2. No reliable object shape

What “object shape” means

- Length, width, height, contours of an object.

Radar case

- Radar reflections depend on **metal surfaces and angles**.
- It returns **strong points**, not surface outlines.

Result

- A car, truck, or guardrail can look **similar** in radar.

- Shape cannot be reconstructed accurately.

LiDAR contrast (for intuition)

- LiDAR hits many surface points → clear shape.
-

3. Ghost targets due to multipath

What is multipath?

- Radar waves **bounce multiple times** before returning.

How ghost objects appear

1. Radar signal hits Car A
2. Reflects to road / wall
3. Reflects again to radar
4. Radar thinks the object is **somewhere else**

Result

- False objects appear that **do not exist**.
- These are called **ghost targets** or **phantom objects**.

Why dangerous

- ADAS may brake or warn unnecessarily.
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4. Hard to localize obstacles precisely

What “localize” means

- Knowing **exact position**:
 - distance
 - left/right offset
 - exact lane position

Radar case

- Distance: good
- Velocity: very good
- Lateral position (x-axis): **poor**

Result

- Radar may say:

“Object at 40 m”

- But cannot say clearly:

"Is it in my lane or the next lane?"

ADAS impact

- Lane-level decisions become unreliable.

Final intuition summary

Problem	Why it happens in radar
Poor angular resolution	Wide beams
No clear shape	Sparse reflections
Ghost objects	Multipath reflections
Poor localization	Weak lateral accuracy

Radar answers: "Something is there and moving."

LiDAR answers: "What it is, where it is, and what shape it has."

You said:

explain these

Correct version:

Explain these points.



adas all level

ADAS levels (SAE Levels 0-5) explained step by step

Level 0 — No Automation

- **Who controls?** Human (100%)
- **What the system does?**
 - Only warnings, no control.
- **Examples:**
 - Forward collision warning
 - Lane departure warning
- **Key point:**
 - System **never** steers, brakes, or accelerates.

Level 1 — Driver Assistance

- **Who controls?** Human
 - **What the system does?**
 - Assists in **one task only**.
 - **Examples:**
 - Adaptive Cruise Control (speed only)
 - Lane Keeping Assist (steering only)
 - **Key point:**
 - Either steering **or** speed, not both.
-

Level 2 — Partial Automation

- **Who controls?** Human (must monitor continuously)
 - **What the system does?**
 - Controls **steering + acceleration + braking** together.
 - **Examples:**
 - Tesla Autopilot
 - GM Super Cruise
 - **Key point:**
 - Hands-off sometimes, **eyes-on always**.
 - Driver is legally responsible.
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Level 3 — Conditional Automation

- **Who controls?** System (under conditions)
 - **What the system does?**
 - Full driving in **specific scenarios**.
 - **Examples:**
 - Highway traffic jam pilot
 - **Key point:**
 - Driver can relax **but must take over when requested**.
 - **Big challenge:**
 - Safe and fast human handover.
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Level 4 — High Automation

- **Who controls?** System

- **What the system does?**
 - Drives fully **within a defined area or condition**.
- **Examples:**
 - Robo-taxis in geo-fenced cities
 - Autonomous shuttles
- **Key point:**
 - No driver needed **inside the operating domain**.

Level 5 — Full Automation

- **Who controls?** System (always)
- **What the system does?**
 - Drives **anywhere, anytime**, like a human.
- **Examples:**
 - None in real-world deployment yet.
- **Key point:**
 - No steering wheel or pedals required.

Quick comparison table

Level	Automation	Driver attention
0	None	Always
1	Single assist	Always
2	Steering + speed	Always
3	Conditional	On request
4	High (limited area)	Not required
5	Full	Not required

Important ADAS insight

- **ADAS usually refers to Level 1 & Level 2.**
- Level 3-5 are **autonomous driving**, not just assistance.

You said:

adas all level

Correct version:

Explain all ADAS levels.



how does yolo tells depth?

YOLO by itself does NOT know depth.

It predicts **2D bounding boxes only**. Depth is added using extra assumptions or extra models.

Below are **all the common ways YOLO is used to get depth**, explained step by step.

1 YOLO + Camera Geometry (size-based depth)

Idea:

If you know the **real-world size** of an object and its **size in the image**, you can estimate depth.

Step-by-step

1. YOLO detects an object → bounding box (w_{px}, h_{px})
2. You assume real height H (e.g., car ≈ 1.5 m)
3. Use pinhole camera model:

$$Z = \frac{f \cdot H}{h_{px}}$$

Where:

- Z = depth (distance)
- f = focal length (from camera calibration)
- H = real object height
- h_{px} = height in pixels

Limitations

- Assumes fixed object size ✗
- Bad for different car models, people, poses ✗
- Rough estimate only

2 YOLO + Monocular Depth Estimation Network

Most common modern approach

Pipeline

1. YOLO → detects **where** the object is

2. Depth network (MiDaS, DPT, MonoDepth2) → predicts **depth map**
3. Crop depth map using YOLO bounding box
4. Average / minimum depth inside the box → object depth

Why this works

- Depth network learns **perspective cues**:
 - texture gradients
 - object scaling
 - occlusion
 - horizon position

Important truth

- This depth is **relative**, not absolute
- Needs scaling or calibration

3 YOLO + Stereo Camera

Human-like depth perception

Step-by-step

1. Left image → YOLO detects object
2. Right image → YOLO detects same object
3. Compute disparity d
4. Depth formula:

$$Z = \frac{f \cdot B}{d}$$

Where:

- B = baseline (distance between cameras)

Pros

- True geometric depth 
- Accurate in short-mid range

Cons

- Needs two cameras
- Sensitive to lighting and texture

4 YOLO + LiDAR (Best for ADAS)

Step-by-step

1. LiDAR → gives true 3D points
2. Project LiDAR points to image
3. YOLO detects object in image
4. Associate LiDAR points inside bounding box
5. Median / min depth = object distance

Result

- **Accurate, metric depth**
- Robust and reliable

5 YOLO trained to regress depth (research)

Some YOLO variants directly predict:

$$(x, y, w, h, Z)$$

Reality check

- Works only on specific datasets
- Generalization is poor
- Not used in safety-critical systems

Final Truth (Very Important)

Method	True depth?	Reliability
YOLO alone	✗ No	✗
YOLO + geometry	Approx	⚠
YOLO + mono depth	Relative	⚠
YOLO + stereo	✓ Yes	✓
YOLO + LiDAR	✓ Best	★★★★

One-line conclusion

YOLO never “knows” depth — depth always comes from geometry, learning, or another sensor.

You said:

how does yolo tells depth?

Correct version:

How does YOLO estimate depth?



so it predicts H then calculates depth?

No. YOLO does NOT predict H .

Let's be very clear and precise.

Case 1: YOLO + size-based geometry (simple method)

YOLO predicts **only**:

$$(x, y, w_{px}, h_{px})$$

- $h_{px} \rightarrow$ **pixel height**
- $H \rightarrow$ **assumed / known real-world height**

$$Z = \frac{f \cdot H}{h_{px}}$$

Important truth

- H is **NOT** predicted
- H is **manually assumed** (e.g., car = 1.5 m, person = 1.7 m)

✗ YOLO never learns H

✗ YOLO never outputs H

Case 2: YOLO + monocular depth network

- YOLO $\rightarrow$ object location
- Depth network $\rightarrow$ **depth per pixel**

Here:

- No H
- No object-size assumption
- Depth is inferred from **learned perspective cues**

But depth is **relative**, not absolute.

Case 3: YOLO + stereo or LiDAR (real systems)

- Geometry gives depth directly
- YOLO is used only for **association**
- No H involved

Why predicting H is a bad idea

- Object heights vary (SUV vs sedan, adult vs child)
- Pose changes (sitting, bending)
- Occlusion breaks estimation

So **predicting H** is unstable and unsafe.

Final crisp answer

- ✗ YOLO does **not** predict H
- ✓ YOLO predicts **2D bounding boxes**
- ✓ Depth comes from **assumptions, geometry, learning, or other sensors**

You said:

so it predicts H then calculates depth?

Correct version:

So, does it predict H and then calculate depth?



how it will calculate h_{pixel}

YOLO does NOT calculate h_{pixel} .

It directly predicts it.

Let's go step by step.

What h_{pixel} means

h_{pixel} = height of the bounding box in image pixels

How YOLO gets h_{pixel}

Step 1: YOLO outputs bounding box parameters

YOLO predicts for each object:

$$(x_c, y_c, w, h)$$

- $x_c, y_c \rightarrow$ center of box (normalized or grid-based)
- $w, h \rightarrow$ **width and height of box (normalized)**

These are **learned values**, not computed from geometry.

Step 2: Convert to pixel values

If image height = H_{img} pixels:

$$h_{pixel} = h \times H_{img}$$

This is a **simple scaling**, not estimation.

Where does YOLO learn this from?

Training phase

- Ground truth labels contain **exact bounding box heights in pixels**
- Loss function penalizes error between:

$$h_{pred} \text{ vs } h_{gt}$$

YOLO learns to predict box height by:

- edges
 - texture
 - object extent
 - context
-

Important clarification

- YOLO **does not use camera parameters**
- YOLO **does not use depth**
- YOLO **does not calculate height physically**

It only predicts **2D box size**.

Why this works

Objects appear smaller when far → YOLO learns this **implicitly**
But it never converts this into depth.

One-line summary

$$h_{pixel} = \text{YOLO-predicted bounding box height}$$

You said:

how it will calculate h_{pixel}

Correct version:

How does it calculate h_{pixel} ?



then how f?

f is **NOT** predicted by YOLO.
It comes from **camera calibration**.

What f actually is

- f = **focal length in pixels**
- It is a **camera property**, not a learned value.

How f is obtained (step by step)

Step 1: Camera calibration

You calibrate the camera using:

- checkerboard images
- known 3D-2D correspondences

This gives the **camera intrinsic matrix**:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Here:

- f_x, f_y = focal lengths in pixels

- c_x, c_y = principal point

Step 2: Choose focal length

For depth estimation using object height:

$$f \approx f_y$$

because height is measured in the **vertical direction**.

Why focal length is in pixels

Physical focal length (mm) is converted to pixels:

$$f_{pixel} = \frac{f_{mm}}{\text{sensor size (mm)}} \times \text{image size (pixels)}$$

This conversion is handled automatically during calibration.

Final depth equation (full picture)

$$Z = \frac{f \cdot H}{h_{pixel}}$$

- $f \rightarrow$ from **camera calibration**
- $H \rightarrow$ **assumed real-world height**
- $h_{pixel} \rightarrow$ **YOLO-predicted bounding box height**

YOLO is involved **only** in h_{pixel} .

Important truth (ADAS reality)

- Real ADAS systems **do not rely on this formula**
- They use **LiDAR, radar, or stereo**
- This equation is mainly for **academic or rough estimation**

One-line summary

f comes from camera calibration, not from YOLO

You said:
then how f ?

Correct version:

Then how is f obtained?



tesla cars uses adas 3?

No. Tesla cars do NOT use Level 3 ADAS.

They operate at **SAE Level 2**.

Why Tesla is Level 2, not Level 3

1 Driver responsibility

- In Level 3, the **system is legally responsible** while active.
- In Tesla:
 - Driver must keep attention
 - Driver must intervene anytime
- That is **Level 2 by definition**.

2 Monitoring requirement

- Tesla requires:
 - Steering wheel torque (earlier)
 - Camera-based driver monitoring (now)
- If driver is inattentive → warnings → disengagement
 - 👉 This means **continuous human supervision** → Level 2.

3 No true handover logic

- Level 3 systems:
 - Say: *"I cannot handle this, take over"*
 - Give **guaranteed safe time** for takeover
- Tesla:
 - Does **not** guarantee safe fallback
 - Expects driver to be ready **always**

4 Legal classification

- Tesla officially markets it as:
 - “Autopilot”
 - “Full Self-Driving (Supervised)”
- The word “Supervised” itself implies **Level 2**.

Comparison with real Level 3 system

Feature	Tesla FSD	Mercedes Drive Pilot
SAE Level	2	3
Driver attention	Required	Not required (conditions)
Legal liability	Driver	Manufacturer
Geo-fenced	No	Yes
Speed limited	No	Yes

Important insight

Tesla avoids Level 3 **on purpose**:

- Level 3 requires legal liability
- Requires extremely strict safety guarantees
- Tesla prefers faster iteration under Level 2 rules

Final answer (one line)

Tesla cars use SAE Level 2 ADAS, not Level 3.

You said:

tesla cars uses adas 3?

Correct version:

Do Tesla cars use Level 3 ADAS?